# Lie Algebras

**Def** A <u>Lie algebra</u> over $k$ is a vector space $\mathfrak{g}$ over $k$ endowed with a bilinear operation

$$[\,,\,] : \mathfrak{g} \otimes \mathfrak{g} \longrightarrow \mathfrak{g}$$

satisfying:

- [skew-symmetry] $\quad [x,y] = -[y,x] \quad \forall\, x,y \in \mathfrak{g}$
- [Jacobi identity] $\quad [x,[y,z]] + [y,[z,x]] + [z,[x,y]] = 0$

$\forall\, x,y,z \in \mathfrak{g}$.

**Def** A Lie algebra is <u>commutative</u> if $[\,,\,]=0$

<u>Idea</u> To every Lie algebraic group we will associate a Lie algebra. We can understand the structure and representation theory of the group through the Lie algebra.

$$\left\{ \begin{array}{c} \text{Lie algebraic} \\ \text{groups} \end{array} \right\} \longrightarrow \left\{ \begin{array}{c} \text{finite-dimensional} \\ \text{Lie algebras} \end{array} \right\}$$

$$\begin{array}{c} \text{geometric} \\ \text{methods} \end{array} \longleftrightarrow \text{algebraic methods}$$

[the map is roughly finite-to-one].

**Lemma** Let $E$ be an associative algebra over $k$. We can endow $E$ with the structure of a Lie algebra by:

$$[\,,\,] \quad E \otimes E \longrightarrow E$$

$$x, y \longmapsto xy - yx \qquad \text{"commutator"}$$

So $E$ is commutative as an associative algebra iff it is commutative as a Lie algebra.

pf Easy verification. $\square$

**Aside** This is a forgetful functor $\left\{\begin{matrix}\text{assoc.}\\ \text{algebras}\end{matrix}\right\} \longrightarrow \{\text{Lie algebras}\}$

We'll see the adjoint later $\downarrow$

**Examples**

(1) Let $V$ be a vector space $/k$. The endomorphism algebra $E = \mathrm{End}(V)$ has a Lie algebra structure via the commutator. We will write $\mathfrak{gl}(V)$ instead of $\mathrm{End}(V)$ when we want to [illegible] as a Lie algebra.

(2) Let $V$ be as above. The subspace

$$\{ f \in \mathrm{End}(V) \quad tr(f) = 0 \}$$

is a Lie subalgebra of $\mathrm{End}(V)$ $(= \mathfrak{gl}(V))$ since $tr(fg) = tr(gf)$

for $\mathfrak{sl}$ finite $\mathrm{End}(V)$.

(Note: trivialities automorphisms are defined independent of any basis.

They are not an associative subalgebra of $\mathrm{End}(V)$. )

We denote this Lie subalgebra by $\mathfrak{sl}(V)$

(3) Let $R$ be a commutative ring and $A$ an $R$-algebra, also commutative.

Then $\mathrm{Der}_R(A,A)$ is a Lie subalgebra of $\mathrm{End}_R(A)$.

Check: Given $D_1, D_2 \in \mathrm{Der}_R(A,A)$, show that

$D_1 \circ D_2 - D_2 \circ D_1$ is again a derivation.

Note: $\mathrm{Der}_R(A,A)$ is not an associative subalgebra of $\mathrm{End}_R(A)$).

(4) Let $A$ be as in (3). Suppose a group $G$ acts on $A$ by $R$-algebra automorphisms. So

$$\rho : G \longrightarrow GL(A)$$

where $\rho(g) : A \to A$ an $R$-algebra automorphism.

Then $G$ acts on $\mathrm{End}_R(A)$ by:

$$(g \cdot f)(a) = g\, f(g^{-1} a)$$

Claim: $\mathrm{Der}_R(A,A)$ is stable under the $G$-action

$$\left[\text{i.e. } D \in \mathrm{Der}_R(A,A) \Rightarrow g \cdot D \in \mathrm{Der}_R(A,A)\right]$$

$\forall g \in G$

(b) The invariants

$$\mathrm{End}_R(A)^G = \{ f \in \mathrm{End}_R(A) : \quad g \cdot f = f \quad \forall g \in G \}$$

$$\mathrm{Der}_R(A,A)^G = \{ D \in \mathrm{Der}_R(A,A) \quad g \cdot D = D \quad \forall g \in G \}$$

$$= \mathrm{Der}_R(A,A) \cap \mathrm{End}_R(A)^G$$

are Lie subalgebras (Check this!)

**Def** Let $\mathfrak{g}$, $\mathfrak{h}$ be Lie algebras. A <u>morphism of Lie algebras</u> is a linear map $\phi : \mathfrak{g} \longrightarrow \mathfrak{h}$ s.t.

$$\phi([x,y]) = [\phi(x), \phi(y)] \qquad \forall x, y \in \mathfrak{g}.$$

A <u>representation of a Lie algebra</u> $\mathfrak{g}$ is a morphism of Lie algebras

$$\mathfrak{g} \longrightarrow \mathfrak{gl}(V) \qquad \text{for some v.s. } V.$$

**Def** Let $X$ be an affine variety. The Lie algebra of vector fields on $X$ is defined as

$$\mathcal{T}_X := \mathrm{Der}_k(k[X], k[X])$$

The space of 1-forms on $X$ is

$$\Omega_X = \Omega_{k[X]}$$

<u>Note</u>. For any $x \in X$, we have that

$$T_x X = \mathrm{Der}(k[X], k_x) = \mathcal{T}_X \otimes_{k[X]} k_x$$

$$T_x^* X = \Omega_{k[X]} \otimes_{k[X]} k_x = \Omega_X \otimes_{k[X]} k_x$$

Thus we have maps

$$\mathcal{T}_X \longrightarrow T_x X \qquad \Omega_X \longrightarrow T_x^* X$$

Also $T_x^* X = (T_x X)^*$ as vector spaces

Now suppose a linear algebraic group $G$ acts on an affine variety $X$.

We have

Action map $\quad G \times X \longrightarrow X$

Induces $\quad k[X] \longrightarrow k[G] \otimes k[X]$ (coaction)

$$f \longmapsto [g, x \longmapsto f(gx)]$$

Equivalently $G$ on $k[X]$

$$G \longrightarrow GL(k[X])$$

$$g \longmapsto \xi_g \quad [f \longmapsto [x \longmapsto f(g^{-1}x)]]$$

Is algebra automorphism of $k[X]$

<u>Note</u>. $\xi_g$ induces $\quad a_g : X \longrightarrow X$

$$x \longmapsto gx$$

**Def** The Lie algebra of $G$-equivariant vector fields on $X$ is defined as

$$(\mathcal{T}_X)^G = \mathrm{Der}(k[X], k[X])^G \subseteq \mathcal{T}_X$$

Explicitly,

$$(\mathcal{T}_X)^G = \{ \partial \in \mathrm{Der}(k[X], k[X]) : \theta \circ \rho_g = \rho_g \circ \theta \text{ for all } g \in G \}$$

**Ex** $G = \mathbb{G}_m \curvearrowright X = \mathbb{C}$ by scalar multiplication $x \cdot a = \lambda a$ for $\lambda \in \mathbb{G}_m$, $a \in k$

Then the action of $\lambda \in \mathbb{G}_m$ on $k[X] = k[T]$ is given by:

$$\rho_\lambda \quad k[T] \longrightarrow k[T]$$
$$T \longmapsto \lambda^{-1} T$$

Observe that $\rho_\lambda(f) = \lambda^{-n} f$ $\forall \lambda \in \mathbb{G}_m$ iff $f$ is homogeneous of degree $n$.

Now any derivation $k[T] \longrightarrow k[T]$ has the form $V \frac{\partial}{\partial T}$ for some $V \in k[T]$. We compute:

$D = V \frac{\partial}{\partial T}$ is in $\mathrm{Der}(k[X], k[X])^G$

iff $\rho_\lambda \circ D = D \circ \rho_\lambda$ $\forall \lambda \in \mathbb{G}_m$

iff $D(\lambda^{-1} T) = \rho_\lambda(D(T))$ $\forall \lambda \in \mathbb{G}_m$

iff $\quad v \frac{\partial}{\partial T}(\lambda T) = \rho_\lambda\left(v \frac{\partial}{\partial T}(T)\right) \qquad \forall \lambda \in \mathbb{G}_m$

iff $\quad \lambda v = \rho_\lambda(v) \qquad \forall \lambda \in \mathbb{G}_m$

iff $\quad v \in k[T]$ is homogeneous of degree $1$

iff $\quad v \in k\{T\} = k[T]$

iff $\quad v \in k\{T \frac{\partial}{\partial T}\}$

Thus, the space of $G$-equivariant vector fields is the span of $\overbrace{T \cdot \frac{\partial}{\partial T}}^{\text{Euler operator}}$ (one-dimensional).

<u>Extension</u>. Compute

$$(\mathcal{T}_{k^n})^{\mathbb{G}_m} \subseteq \mathcal{T}_{k^n} = k[T_1, \dots, T_n]\left\{\frac{\partial}{\partial T_1}, \dots, \frac{\partial}{\partial T_n}\right\}.$$

where $\mathbb{G}_m \curvearrowright k^n$ by scalar multiplication. $\boxtimes$

$\boxed{\text{Prop}}$ Let $G$ be a linear algebraic group and $X$ an affine $G$-variety.

(1) Let $\partial \in \mathcal{T}_X$. Then $\partial \in (\mathcal{T}_X)^G$ iff

$$d_x(\rho_g)(\partial_x) = \partial_{gx} \qquad \forall g \in G,\ x \in X$$

(2) $\mathcal{T}_X^G$ is a subalgebra of $\mathcal{T}_X$.

② If $X$ is a principal homogeneous space (torsor) for $G$, then, for any $x \in X$, the map

$$(\tilde{\tau}_X)^G \longrightarrow T_x X$$

$$\theta \longmapsto \theta_x \quad (= \pi_x \circ \theta, \text{ where } \pi_x : k[X] \to k_x)$$

is an isomorphism of vector spaces.

Pf. ① Let $\theta \in \tilde{\tau}_X$ be a vector field. Then

$$\theta \in (\tilde{\tau}_X)^G \quad \text{iff} \quad \rho_g \circ \theta = \theta \circ \rho_g \quad \forall g \in G$$

$$\text{iff} \quad a_g^* \circ \theta = \theta \circ a_g^* \quad \forall g \in G \quad (\text{or } g^{-1} \in G)$$

$$\text{iff} \quad [(a_g^* \circ \theta)(f)](x) = [(\theta \circ a_g^*)(f)](x) \quad \forall g \in G, \ \forall f \in k[X], \ \forall x \in X$$

$$\text{iff} \quad \theta(f)(gx) = d_x(a_g)(\theta_x)(f) \quad \forall g \in G, \ \forall f \in k[X], \ \forall x \in X$$

$$\text{iff} \quad \theta_{gx}(f) = d_x(a_g)(\theta_x)(f) \quad \sim\text{"}\sim$$

$$\text{iff} \quad \theta_{gx} = d_x(a_g)(\theta_x) \quad \forall g \in G, \ x \in X.$$

② Immediate from the fact that $\mathrm{Der}(k[X], k[X])^G$ is a Lie subspace of $\mathrm{Der}(k[X], k[X])$

(3) The choice of $x \in X$ gives an isomorphism of varieties

$$G \longrightarrow X$$
$$g \longmapsto g\,x$$

Let $\psi : k[X] \longrightarrow k[G]$ be the induced isomorphism.

Define:

$$s : T_x X \longrightarrow \mathrm{End}_k(k[X])$$

as follows, for $\xi \in T_x X = \mathrm{Der}(k[X], k_x)$:

$$s(\xi) = \left[ k[X] \xrightarrow{a^*} k[G] \otimes k[X] \xrightarrow{\psi^{-1} \otimes 1} k[X] \otimes k[X] \xrightarrow{\xi \otimes 1} k_x \otimes k[X] \to k[X] \right]$$

More explicitly:

$$\left[ s(\xi)(f) \right](g\,x) = \left[ d_x(a_g)(\xi) \right](f) \qquad (*)$$

<u>To check:</u>

- $s(\xi)$ is a derivation $\forall\, \xi \in T_x X$
- $s(\xi)$ is $G$-equivariant $\forall\, \xi \in T_x X$. $\square$

<u>Remark</u>: Part (3) is true whenever $X = G/H$ for a normal subgroup $H$ of $G$, and essentially the same proof works (with some modifications to check that $s$ is well-defined)

Let $G$ be a Lie algebra group. Recall the actions of left and right translation

$$\ell_g : G \longrightarrow G \qquad\qquad r_g : G \longrightarrow G$$
$$x \longmapsto gx \qquad\qquad x \longmapsto xg^{-1}$$

Denote the Lie algebra of left- and right-invariant vector fields on $G$ by

$$\mathrm{Lie}(G)_\ell = \left(\mathfrak{X}_G\right)^{\text{left-invariant}} = \{\theta \in \mathfrak{X}_G : \ell_g^* \circ \theta = \theta \circ \ell_g^*\}$$

$$\mathrm{Lie}(G)_r = \left(\mathfrak{X}_G\right)^{\text{right-invariant}} = \{\theta \in \mathfrak{X}_G : r_g^* \circ \theta = \theta \circ r_g^*\}$$

**Lemma** Let $S : \mathfrak{b}(G) \to \mathfrak{b}(G)$ be the antipode. The map

$$\mathrm{Lie}(G)_r \longrightarrow \mathrm{Lie}(G)_\ell$$
$$\theta \longmapsto S \circ \theta \circ S$$

is an isomorphism of Lie algebras.

*proof* Recall that $S = i^*$ where $i : G \longrightarrow G$ is the "inverse" $x \mapsto x^{-1}$. Note that the following diagrams commute:

$$\begin{tikzcd} G \arrow[r, "r_g"] \arrow[d, "i"'] & G \arrow[d, "i"] \\ G \arrow[r, "\ell_g"] & G \end{tikzcd} \qquad\qquad \begin{tikzcd} G \arrow[r, "\ell_g"] \arrow[d, "i"'] & G \arrow[d, "i"] \\ G \arrow[r, "r_g"] & G \end{tikzcd}$$

To check

- $\varphi$ is well-defined:

$$S \circ \partial \circ S \circ l_S^* = S \circ \partial \circ i^* \circ l_S^* = S \circ \partial \circ (l_S \circ i)^*$$

$$= S \circ \partial \circ (i \circ r_S)^* = S \circ \partial \circ r_S^* \circ i^*$$

$$= S \circ r_S^* \circ \partial \circ S = l_S^* \circ S \circ \partial \circ S .$$

- $\varphi$ is a map of Lie algebras
  (use $i^2 = id_G$ so $S^2 = id_{k[G]}$)

$\square$

- $\varphi$ is an isomorphism.

Here's another way to think about this algebra.

Find the coproduct map

$$\Delta \quad k[G] \longrightarrow k[G] \otimes k[G]$$

and counit $\eta = ev_1 : k[G] \longrightarrow k$.

$\boxed{\text{Lemma}}$ The dual vector space $k[G]^* = Hom_k(k[G], k)$ of $k[G]$ becomes an associative algebra structure given by

$$k[G]^* \otimes k[G]^* \longrightarrow k[G]^*$$

$$\xi \otimes \xi' \longmapsto \xi \cdot \xi' = (\xi \otimes \xi') \circ \Delta$$

with unit $\eta \in k[G]^*$.

<u>proof</u> Follows from the coassociativity of $\Delta$ and the counit axiom $\square$

As we've seen we get a Lie algebra structure on $k[G]^*$ via the commutator

$$[\xi, \xi'] = \xi \cdot \xi' - \xi' \cdot \xi \qquad \text{for } \xi, \xi' \in k[G]^*.$$

$\boxed{\text{Lemma}}$ The subspace $T_1 G = \mathrm{Der}(k[G], k) \subseteq k[G]^*$ is a Lie subalgebra of $k[G]^*$

<u>proof</u> We use Sweedler notation $\Delta(a) = a_1 \otimes a_2$ for $a \in k[G]$ so the summation is implicit.

For $\xi, \xi' \in T_1 G$ and $a, b \in k[G]$ we have:

- $\xi \cdot \xi'(a) = \xi(a_1)\, \xi'(a_2)$

- $\xi \cdot \xi'(ab) = \xi(a_1 b_1)\, \xi'(a_2 b_2)$

$$= [a_1 \cdot \xi(b_1) + b_1 \cdot \xi(a_1)][a_2 \cdot \xi'(b_2) + b_2\, \xi'(a_2)]$$

$$= \dots$$

$$= \eta(a)(\xi \cdot \xi'(b)) + \eta(b)(\xi \cdot \xi'(a))$$
$$+ \xi(a)\, \xi'(b) + \xi(b)\, \xi'(a)$$

Using the facts that

- $\eta(a_1 a_2) = \eta(a)$
- $\eta(a_1)a_2 = a = a_1 \eta(a_2)$

) and similarly for $b$

- $\xi$ and $\xi'$ are $k$-linear.

It then follows easily that

$$[\xi, \xi'](ab) = a\left([\xi, \xi'](b)\right) + b\left([\xi, \xi'](a)\right). \quad \square$$

<u>Upshot</u>: we get a Lie algebra structure on $T_1 G$.
We write $\mathfrak{g}$ for $T_1 G$ equipped with this Lie algebra structure.

$\boxed{\text{Proposition}}$ The map

$$\begin{aligned} \mathrm{Lie}(G)_e &\longrightarrow \mathfrak{g} \\ \partial &\longmapsto \eta \circ \partial \end{aligned}$$

defines an isomorphism of Lie algebras.

<u>proof</u> Homework exercise (check website). $\square$

First step is to show that

$$\begin{tikzcd} k[\varepsilon] \arrow[r, "\partial"] \arrow[d, "\Delta"'] & k[\varepsilon] \arrow[d, "\Delta"] \\ k[\varepsilon]\otimes k[\varepsilon] \arrow[r, "1\otimes\partial"] & k[\varepsilon]\otimes k[\varepsilon] \end{tikzcd}$$

commutes for any $\partial \in \mathrm{Lie}(G)_e$

<u>Remark</u> In summary, we have isomorphisms of Lie algebras

$$\begin{tikzcd}
\mathrm{Lie}(G)_\ell \arrow[rr, "\theta \mapsto S\circ\theta\circ S"] \arrow[dr, "\theta \mapsto \eta\circ\theta"'] & & \mathrm{Lie}(G)_r \arrow[dl, "\theta \mapsto -\eta\circ\theta"] \\
& \mathfrak{g} &
\end{tikzcd}$$

[<u>Note</u> Any Lie algebra is isomorphic to its opposite via $x \mapsto -x$ ]

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$\boxed{\text{Question}}$ How can we describe $\mathfrak{g}$ in terms of generators and relations?

$\boxed{\text{(Partial) Answer}}$. Since any Lie algebra group embeds into $GL_n$ the first step is to calculate $T_1 GL_n$. Since $GL_n \hookrightarrow Mat_{n,n}$ is an open subscheme, we have a vector space isomorphism

$$T_1 GL_n \xrightarrow{\sim} T_1 Mat_{n,n}$$

<u>Claim</u> Taking pointwise matrix multiplication gives a coalgebra structure on $k[Mat_{n,n}]$

$$\Delta : k[Mat_{n,n}] \longrightarrow k[Mat_{n,n}] \otimes k[Mat_{n,n}]$$

$$\varepsilon : k[Mat_{n,n}] \longrightarrow k$$

(Note unfortunately, this is only a coalgebra, not a Hopf algebra)

So we have an induced algebra structure on $k[Mat_{n,n}]^*$:

$$\Delta^* : k[Mat_{n,n}]^* \otimes k[Mat_{n,n}]^* \longrightarrow k[Mat_{n,n}]^*$$

Now we claim that

(1) $T_1 Mat_{n,n}$ is a Lie subalgebra.

(2) The homomorphism $(*)$ is one of Lie algebras.

Hence we reduce the problem to understanding $T_1 Mat_{n,n}$.

<u>Notation</u> $k[Mat_{n,n}] = k[T_{ij} : 1 \le i,j \le n]$

$\boxed{\text{Claim}}$ We have an injective algebra homomorphism

$$Mat_{n,n} \longrightarrow k[Mat_{n,n}]^*$$

$$E_{ij} \longmapsto \xi_{ij} := \eta \circ \frac{\partial}{\partial T_{ij}}$$

where $\{E_{ij}\}$ are the elementary matrices. The image of this map is $T_1 Mat_{n,n}$.

To see that this is an algebra homomorphism, first recall the elementary matrix multiplication according to the rule:

$$E_{ij} E_{rs} = \delta_{jr} E_{is}$$

So we have

$$(\xi_{ij} \cdot \xi_{rs})(T_{uv}) = (\xi_{ij} \otimes \xi_{rs})\left(\sum_{w=1}^{n} T_{uw} \otimes T_{wv}\right)$$

$$= \sum_{w} \xi_{ij}(T_{uw})\, \xi_{rs}(T_{wv}) = \begin{cases} 1 & \text{if } i=u \text{ and } s=v \\ & \text{and } j=r \\ 0 & \text{otherwise} \end{cases}$$

$$= \delta_{jr}\, \xi_{is}(T_{uv})$$

The proof is done. We know from previous discussions that $T_e \mathrm{Mat}_{n\times n} = \mathrm{Der}(\mathbb{R}[\mathrm{Mat}_{n\times n}], k_1)$ is spanned by $\{\frac{\partial}{\partial T_{ij}}\}$

for $1 \le i,j \le n$. $\square$

<u>Conclusion</u>

We identify the Lie algebra of $GL_n$ with the Lie algebra corresponding to the associative algebra of $n$ by $n$ matrices

$$\mathfrak{gl}_n = \left\{ \begin{array}{l} n \text{ by } n \text{ matrices with the} \\ \text{bracket given by the commutator} \end{array} \right\}$$

of matrices

The Lie algebra of a general Linear Algebraic group can be identified with a Lie subalgebra of $\mathfrak{gl}_n$, but in general does not arise from an associative algebra

**Ex** (1) $\mathfrak{sl}_2 = \{x \in \mathfrak{gl}_2 : \mathrm{trace}(x) = 0\}$

$$= \mathrm{span}_k\left\{ E = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},\ F = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix},\ H = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right\}$$

satisfies

$$[H,E] = 2E \qquad [H,F] = -2F \qquad [E,F] = H$$